SESI ARMADILLOS - PROJECT: SMILE TRIP

School SESI from Tatuí - SP // Brazil



Ana Hanf: Electrical Mechanics
EIGHTY

Davi Lucca: Programmer
SCIENTIST TEMPUS





Fernando Favier: Set Designer
TRIBOULET

Matheus Spesia: Mechanics
EDGY FROM EGYPT



PRESENTATION DESCRIPTION:

Chosen Features:

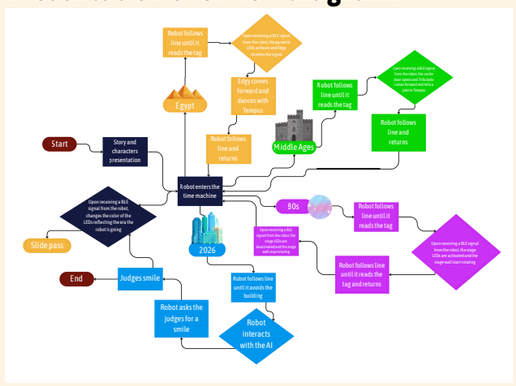
AI RECOGNITION · EXTENDED STAGE · CHRONO MAP · COMMUNICATION ARCHITECTURE

The narrative of our presentation follows the scientist **Tempus** in the year 3026, in a world where people no longer smile and society is devoid of joy. To reclaim the smiles the world has lost, the scientist, along with his assistant, **ALL-IN**, travels to the past in search of them. Throughout the presentation, he observes and shares the true reasons behind the smiles of the past, highlighting their importance for human connection and well-being. This entire performance is the result of a completely authentic and original creation developed by us, ~~without inspiration from or reference to existing works.~~





Presentation overview diagram:



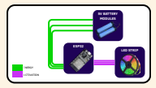
INSTAGRAM · OUR GITHUB · TECHNICAL VIDEO · LOGBOOK

ELECTROMECHANICS:

OUR SCENERY:

Pyramid

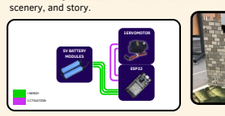
The Pyramid represents our Ancient Egypt setting. It features a WS2812B addressable LED strip controlled by an ESP32. The ESP32 waits for a signal indicating that ALL-IN has reached the Egypt era. Once this signal is received, the lighting effects and the Egyptian robot are activated, starting the scene and synchronizing the scenery with the robot's actions.





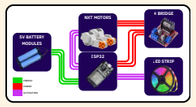
Castle

The Medieval Castle uses a similar system. When ALL-IN leaves the Time Machine and enters the Medieval era, it sends a signal to the castle's ESP32. The ESP32 then activates the motors that open the castle doors, allowing the era's character to emerge and begin the scene. This creates a synchronized transition between the robot, scenery, and story.





80s - System Architecture

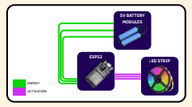




The 1980s stage is designed to be colorful, dynamic, and fun. When ALL-IN leaves the Time Machine and enters this era, an ESP32 receives a BLE signal and activates the entire scene. The system turns on gyro-style lights, controls motors that rotate part of the stage background, and activates WS2812B addressable LED strips, creating a bright and energetic atmosphere inspired by the 1980s.

Time Machine

The Time Machine is one of the main elements of our story. It uses an ESP32 to control its LED system, changing the colors according to each era and the current moment of the performance. In addition, depending on the era, the Time Machine also changes the content displayed on the main screen, creating a synchronized connection between the physical scenery, lighting effects, and the story.





PROGRAMMING LANGUAGES:

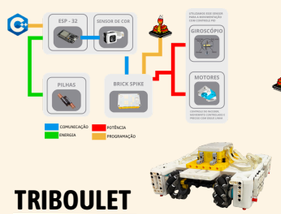
We use five programming languages across the project:

- MicroPython: programs the LEGO SPIKE Prime and EV3 robots. The EV3 uses a micro as a BLE communication bridge.
- Python: runs the smile-recognition AI.
- C++: programs the ESP32 controllers.
- JavaScript: programs the micro during development and testing.
- VBA: automates PowerPoint slide transitions and on-screen content.

OUR ROBOTS:

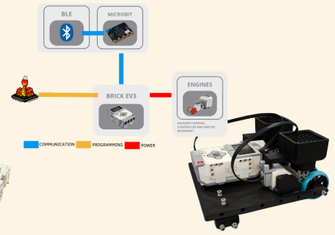
ALL-IN

Our ALL-IN robot uses a precise movement system based on line following with PID control, which helps stabilize its trajectory and improve movement accuracy. The robot is controlled by a LEGO SPIKE Prime running MicroPython. We chose this approach so that, even though ALL-IN is LEGO-based, it can still send and receive BLE signals and communicate with different microcontrollers and peripherals used throughout the project.



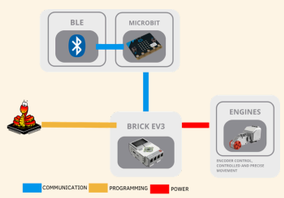
EDGY

The EDGY robot, representing the Ancient Egypt era, has a custom 3D-modeled base and is controlled by a LEGO EV3 running MicroPython. To enable BLE communication, the EV3 is connected to a micro:bit through serial communication. The micro:bit acts as a bridge, receiving and transmitting BLE signals for the EV3. This allows EDGY to remain fully integrated with the performance, being activated only at specific moments of the story.



TRIBOULET

The TRIBOULET robot, representing the Medieval era, uses the same communication architecture as EDGY. It is controlled by a LEGO EV3 running MicroPython, connected to a micro:bit through serial communication. The micro:bit acts as a BLE bridge, allowing TRIBOULET to send and receive signals from the other systems and activate only at the specific moments required by the story.

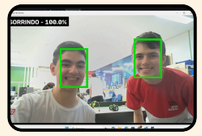


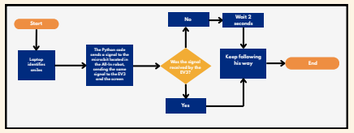


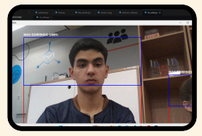
OUR FEATURES:

AI RECOGNITION

AI Recognition uses a custom **YOLO** model trained with around 4,000 labeled images to identify smiling and non-smiling expressions in real time. Running on a Lenovo 2-in-1 laptop with Python and OpenCV, the system analyzes the webcam feed, displays the detected expression and confidence level, and turns smile recognition into an interactive element of the presentation.

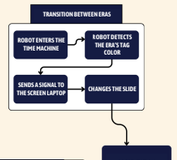


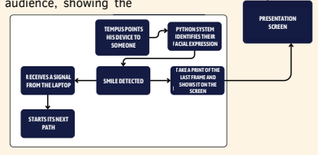




EXTENDED STAGE

Our Extended Stage autonomously reacts to events generated during the performance. When ALL-IN travels between eras, strategic signals trigger synchronized changes on the main screen so the visual content follows the current moment of the story. Once smile recognition is complete, Python saves the last camera frame and sends it to the central notebook, where VBA inserts it into the PowerPoint presentation in real time. Displaying the captured image provides immediate visual feedback to the judges and audience, showing the smile recognized at that exact moment.

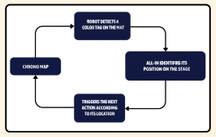


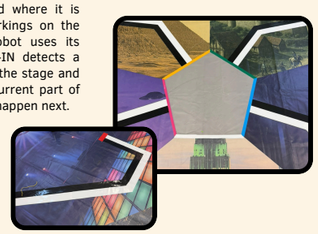




CHRONO MAP

Chrono Map allows ALL-IN to understand where it is during the performance. The colored markings on the floor work as location tags, and the robot uses its sensors to identify each one. When ALL-IN detects a specific color, it recognizes its position on the stage and uses that information to understand the current part of the journey and decide what action should happen next.

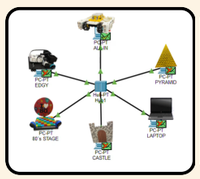


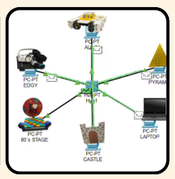


COMMUNICATION ARCHITECTURE

BLE

Our communication architecture connects different microcontrollers, robots, scenery elements, and computer applications through BLE broadcasting. It acts as a bridge between technologies, allowing devices with different hardware and programming languages to exchange signals during the performance. Each part responds to specific signals, coordinating robot movements, scenery effects, and screen updates in real time. This shared communication system allows the individual parts to work together, keeping their actions synchronized with the story.



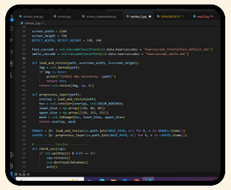


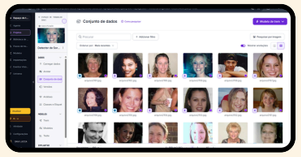


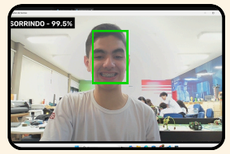
OUR INOVATIONS:

SMILE DEVICE

Our Smile AI was trained in Roboflow using a labeled dataset of smiling and non-smiling faces. We use YOLO for real-time recognition and Python to process the camera feed, display the detected expression, and integrate the result with the rest of the project.

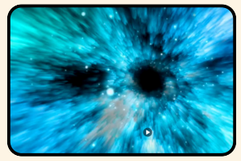






USE OF THE SCREEN

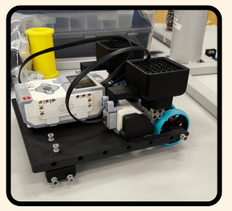
Our Extended Stage connects the physical scenery with the digital presentation. Through BLE signals, ESP32 controllers, and PowerPoint automation, the screen changes together with the robots, lights, and stage elements, creating one synchronized experience between the real and digital parts of the story.





BLE COMMUNICATION

Our BLE communication architecture works like a broadcast network: each device can send signals that are received by all connected systems, while each robot or stage element only reacts to the commands intended for it. This allows the entire performance to stay synchronized without requiring direct one-to-one connections between every device.

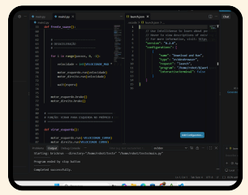


	Voice	Data	Audio	Video	State
Bluetooth ACL / HS	NO	YES	YES	NO	NO
Bluetooth SCO/eSCO	YES	NO	NO	NO	NO
Bluetooth low energy	NO	NO	NO	NO	YES
Wi-Fi	(VoIP)	YES	YES	YES	NO
Wi-Fi Direct	YES	YES	YES	NO	NO
ZigBee	NO	NO	NO	NO	YES

Bluetooth™ Low Energy

EV3DEV AND MICROCONTROLLERS

Our EV3 communication innovation combines EV3Dev/MicroPython, serial communication, and BLE. Because the EV3 does not handle our BLE system directly, a micro:bit works as a bridge, exchanging data with the EV3 through serial while sending and receiving BLE signals from the rest of the project. This allows older EV3 robots to become part of the same modern wireless architecture.



EV3DEV.ORG